

PONTA DELGADA AP, Portugal

WMO: 085120

Lat: 37.7442N

Lon: 25.6969W

Elev: 78

StdP: 100.39

Time Zone: -1.00 (AZR)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.2	10.2	4.1	5.1	11.4	5.1	5.5	11.9	15.1	14.7	13.9	14.5	3.2	0	0.425

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.2	25.3	21.6	24.9	21.3	24.2	20.8	22.9	24.2	22.5	23.8	22.0	23.4	4.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.4	17.2	23.7	22.0	16.8	23.4	21.3	16.1	22.9	68.0	24.5	66.2	23.6	64.4	23.5	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.7	11.0	9.6	7.2	26.9	0.9	0.7	6.6	27.5	6.0	27.9	5.5	28.3	4.9	28.8
			5.6	24.0	0.9	0.9	4.9	24.7	4.4	25.2	3.9	25.7	3.3	26.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.6	14.8	14.3	14.4	15.1	16.6	18.8	21.0	22.2	21.4	19.3	16.9	15.6
	DBStd	3.14	1.61	1.50	1.50	1.43	1.33	1.36	1.28	1.11	1.33	1.69	1.79	1.75
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	648	108	112	121	97	54	11	0	0	9	51	85	
	CDD10.0	2761	150	122	138	153	205	264	342	379	342	288	206	174
	CDD18.3	368	0	0	0	0	2	24	83	121	92	39	6	1
	CDH23.3	378	0	0	0	0	0	2	64	206	97	7	1	0
	CDH26.7	1	0	0	0	0	0	0	0	1	0	0	0	0
Wind	WSAvg	4.5	5.1	5.0	5.4	5.0	4.3	4.0	3.5	3.6	3.8	4.8	4.6	5.4
Precipitation	PrecAvg	1025	129	103	97	71	57	38	28	43	84	114	130	136
	PrecMax	1459	288	250	200	200	192	111	110	141	236	304	282	454
	PrecMin	774	17	24	6	11	2	5	0	6	12	20	10	19
	PrecStd	180	61	57	49	48	43	22	21	34	51	68	64	79
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.6	18.2	18.2	19.8	21.1	23.5	25.9	26.5	25.9	24.2	21.7	20.0
		MCWB	16.8	15.8	15.6	16.1	18.1	19.8	21.5	22.0	22.2	21.3	19.0	18.0
	2%	DB	18.0	17.5	17.7	18.7	20.1	22.6	24.8	25.8	25.1	23.2	20.7	19.1
		MCWB	16.4	15.5	15.4	16.0	17.3	19.3	21.0	21.8	21.7	20.4	18.3	17.4
	5%	DB	17.5	17.1	17.1	18.0	19.5	21.9	24.0	25.1	24.3	22.5	20.0	18.8
		MCWB	16.0	15.3	15.1	15.6	16.9	18.9	20.5	21.4	21.1	20.0	18.0	17.1
	10%	DB	17.1	16.7	16.8	17.2	19.0	21.1	23.2	24.4	23.9	22.0	19.4	18.1
		MCWB	15.7	14.9	14.8	15.1	16.5	18.3	20.1	21.0	20.8	19.7	17.4	16.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.6	17.0	16.9	17.5	19.1	21.3	22.9	23.7	23.6	22.4	20.0	18.7
		MCDB	18.0	17.5	17.5	18.6	20.0	22.4	24.3	25.0	24.8	23.5	20.8	19.4
	2%	WB	17.0	16.4	16.4	16.7	18.4	20.5	22.3	23.1	22.8	21.6	19.4	18.1
		MCDB	17.6	17.1	17.1	17.8	19.4	21.6	23.7	24.5	24.0	22.6	20.2	18.8
	5%	WB	16.6	15.9	15.8	16.4	17.8	19.8	21.7	22.6	22.3	20.7	18.7	17.6
		MCDB	17.3	16.7	16.6	17.4	18.9	20.9	23.1	24.0	23.7	21.9	19.7	18.3
	10%	WB	16.2	15.5	15.5	15.8	17.4	19.3	21.2	22.1	21.7	20.3	18.1	17.1
		MCDB	17.0	16.4	16.4	16.9	18.5	20.6	22.6	23.6	23.2	21.6	19.3	17.9
Mean Daily Temperature Range	5% DB	MDBR	3.6	4.0	4.1	4.3	4.4	4.6	4.9	5.2	4.9	4.4	4.1	3.7
		MCDBR	3.3	3.9	4.1	4.6	4.8	5.1	5.3	5.5	5.2	4.6	4.1	3.4
		MCWBR	3.2	3.5	3.5	3.6	3.5	3.6	3.4	3.5	3.5	3.5	3.6	3.3
	5% WB	MCDBR	2.8	3.2	3.4	3.7	3.8	4.3	4.3	4.4	4.2	3.9	3.8	3.1
		MCWBR	3.2	3.4	3.4	3.3	3.2	3.6	3.4	3.5	3.5	3.7	3.9	3.4
Clear-Sky Solar Irradiance	taub	0.347	0.353	0.375	0.377	0.370	0.372	0.366	0.362	0.361	0.368	0.351	0.352	
	taud	2.532	2.517	2.453	2.455	2.502	2.530	2.556	2.593	2.599	2.557	2.549	2.510	
	Ebn at Noon	835	878	888	903	911	905	908	908	893	851	824	801	
	Edh at Noon	79	90	104	110	107	104	100	95	89	85	77	76	
All-Sky Solar Radiation	RadAvg	2.13	2.98	4.12	5.22	6.14	6.45	6.50	6.14	4.93	3.44	2.39	1.83	
	RadStd	0.16	0.18	0.26	0.40	0.37	0.38	0.32	0.20	0.23	0.19	0.15	0.12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (3 neighbors)	N/A	N/A	N/A	-0.31	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	+0.29	+0.43	N/A	N/A	N/A	N/A

Nomenclature: See separate page